Constantin Seibold, Ph.D.

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Professional Experience

April 2026 - present	**Lecturer**, Department of Computational Linguistics, Heidelberg University
August 2025 - present	**Research Group Leader**, Diagnostic and interventional radiology, University Clinic Heidelberg
May 2023 - Nov. 2024	**Junior Research Group Leader**, Institute for Artificial Intelligence in Medicine, Clinic for Nuclear Medicine, University Medicine Essen
July 2019- Dec. 2022	**Research Assistant**, Computer Vision for Human-Computer Interaction Lab, Karlsruhe Institute of Technology
March 2019- July 2019	**Student Research Assistant**, Computer Vision for Human-Computer Interaction Lab , Karlsruhe Institute of Technology

Education

July 2019 - April 2023	**PhD Computer Science**, Karlsruhe Institute of Technology, Helmholtz Information & Data Science School for Health (HIDSS4Health) Thesis Title: Towards the automatic generation of medical reports in low supervision scenarios Grade: **Summa Cum Laude**, **KIT Doctoral Award** Advisor: Prof. Dr.-Ing. Rainer Stiefelhagen
Oct. 2016 - March 2019	**M. Sc. Computer Science**, Karlsruhe Institute of Technology Thesis Title: Generative Models for Make-Up Style Transfer and Removal Master Thesis Advisor: Dr.-Ing. Muhammed Saquib Sarfraz, Prof. Dr.-Ing. Rainer Stiefelhagen, Institute for Anthropomatics and Robotics
Oct. 2013 - Oct. 2016	**B.Sc. Computer Science**, University of Stuttgart Thesis Title: Improving author co-citation analysis in scientific literature by using citation function classification Bachelor Thesis Advisor: Dr.-Ing. Roman Klinger, Prof. Dr.-Ing. Sebastian Padó, Institute for Natural Language Processing

Funded Research Projects

2026	**Medical Data Scientist Program**, POLARIS: Platform for Optimizing Language-based Assistive Radiology Information Systems, **Role:** PI, **Volume:** 264.000 (100% E14 Position, 3 years)
2025	**Seedfunding des Camilla-und-Georg-Jellinek-Zentrum für Ethik**, Umgang mit unsicherem Wissen bei der Forschung zu krankheitsrelevanten Prädiktoren in der KI-gestützten medizinischen Bildgebung im Rahmen des Lungenkrebsscreenings, **Role:** Support in Grant Application Process, **Volume:** 30.000€
2024	**EU HORIZON JU - Innovative Health Initiative - *HORIZON-JU-IHI-2023-05-02***, Thera4Care - THERANOSTICS ECOSYSTEM FOR PERSONALISED CARE, **Role:** PI, **Sub-Project Volume:** 260.000€ [Link]
2022	**HIDSS4Health**, Using Anatomical Knowledge to Improve Medical Image Analysis, **Role:** Support in Grant Application Process, **Volume:** 130.000€ (50% PhD position, 3 years) [Link]
2022	**HIDSS4Health**, Interactive Annotation of Volumetric Imaging Data Incorporating Report Information, **Role:** Support in Grant Application Process, **Volume:** 130.000€ (50% PhD position, 3 years) [Link]
2019	**HIDSS4Health**, Generating Medical Imaging Reports from 3D Radiological CT Scans using Image Captioning Techniques, **Role:** Team-member, project implementation[Link]

Patents

2024	**Aktenzeichen: 10 2023 112 553.2**, Computerimplementiertes Verfahren zum Trainieren eines maschinellen Lernmodells für die semantische Bildsegmentierung [DPMA]

Awards

2026	**Best Presentation Award**, ECCV - Data Curation and Augmentation in Medical Images Workshop
2024	**2nd Place**, Automated Lesion Segmentation in Whole-Body PET/CT Challenge - autoPET III [paper]
2024	**1st Place (Spanish, Chinese), 3rd Place English**, MEDIQA-M3G Challenge at NAACL-ClinicalNLP [paper]
2023	**Winning Team of the UME - Innovation Contest**, University Medicine Essen
2023	**KIT Doctoral Award**, Karlsruhe Institute of Technology [link]
2022	**Student Travel Award**, Medical Image Computing and Computer Assisted Intervention Society [link]
2022	**Teaching Award - Best Practical Course**, Karlsruhe Institute of Technology, CS Faculty [link]
2019	**Best Industry Paper Award**, British Machine Vision Conference [paper]

Teaching Experience

Theses

Since 2024	**Supervision of Ph.D. Theses**, UME; Students: H. Kalisch
2023-2024	**Co-Supervision of Ph.D. Theses**, UME; Students: L. Heine, A. Brehmer, F. Jonske
2020-2024	**Supervision of Master Theses**, KIT, UME; Students: P. Nguyen, W. Di, P. Albrecht, R. Chlebecec
2020, 2024	**Supervision of Bachelor Theses**, KIT, UME; Students: C. Goos, J. Nasimzada, S. Mahler

Courses

2026	**Projektseminar: Identifying Structure in Data**, University Heidelberg; Lecturer [SS26]
2022	**Deep Learning for Computer Vision I: Basics**, KIT; Teaching Assistant [SS22]
2021-2023	**Deep Learning for Computer Vision II: Advanced Topics**, KIT; Teaching Assistant, Lecturer [WS 22,WS 23]
2021-2022	**Practical Course - Computer Vision for Human Computer Interaction**, KIT; Lecturer/Course Organizer **[Best Practical Course of SS21]** [WS21, SS21, WS22], consistently LQI 100 (”Lehrqualitätsindex”, 100 is the best possible value at KIT).
2020-2021	**Deep Learning for Computer Vision**, KIT; Teaching Assistant [SS20,SS21]

Community Engagement & Outreach

2026-2027	**Marsilius Kolleg**, Young Marsilius Fellow, 5000 Euro Stipend
2026-current	**ELLIS - European Laboratory for Learning and Intelligent Systems**, Member
2026-current	**Camilla-und-Georg-Jellinek-Zentrum für Ethik**, Member
2025	**CVPR Tutorial - Identifying Structure in Data**, Organizer [link]
2023-2024	**Initiating and organizing the Computer-Vision Reading Group at the IKIM**, Organizer, Presenter
2021-2022	**Initiating and organizing the Computer-Vision Reading Group at CV:HCI**, Organizer, Presenter
2022	**MICCAI Workshop - Medical Applications with Disentanglements**, Program Committee

Invited Talks

2026	**Seoul, Korea**, KAIST ICML Panel Discussion on Language Model Reliability & Safety
2025	**#algorithm-meeting @ DeepHealth**, Online Dimensionality Reduction and Clustering Tutorial
2025	**AI, ML and Computer Vision Meetup**, Online (hosted by voxel51) Clustering in Computer Vision: From Theory to Applications
2024	**ENBIS Spring Meeting**, Dortmund, Germany Interpretable AI in Medicine: Generating Radiological Reports with Panoptic Scene Graphs
2024	**Intelligent Sensing and Perception Group**, Stuttgart, Germany The Current state of vision-language models in Radiology

Research Datasets & Software

Public Datasets

2022–2023	**PAXRay / PAXRay++**, First author. 10,021 thoracic CTs, 157 anatomical labels projected to radiograph geometry; enables dense anatomy segmentation without manual annotation. BMVC 2022; extended 2023. [data]
2023	**DAP Atlas**, Co-first author. Anatomical masks for 533 CT scans covering 142 structures, built on AutoPET. ICIP 2024. [paper]
2025	**MedShapeNet**, Co-author. Over 100,000 medical 3D shapes – bones, organs, vessels, surgical instruments – for translating computer-graphics methods to clinical use. [data]
2022	**Synthetic Structured Visual Content (SSVC)**, Co-author. 12,000 images of graphs and flow charts with bounding boxes and graph ground truth; supports machine reading of diagrams and accessibility for visually impaired readers. ICPR 2022.
2021	**Sims4Action**, Co-author. Ten hours of simulated activities of daily living across eight characters, camera angles and environments. IROS 2021.
2019	**Facial Cosmetic Content (FCC)**, Co-author. 18,425 facial images across strong, mild and no makeup, in 256×256 and 512×512 subsets. BMVC 2019 (Best Industry Paper).

Open-Source Software

since 2023	**CXAS – Chest X-Ray Anatomy Segmentation**, Maintainer. PyPI package for fine-grained anatomy segmentation in chest radiographs. 28k downloads, [code]
2024	**Spacewalker**, Senior author. Interactive exploration and annotation of unstructured data. MLVis@EuroVis 2024. [code]
since 2024	**2DAnatomyDatasets**, Maintainer. Curated collection of 2D anatomy-segmentation datasets. [code]
since 2020	**SGL – Self-Guiding Loss**, First author. Reference implementation, ACCV 2020. [code]

Further Education

2023	**Workshop KHYS Basics of Leadership in Science 2023**, engaged with essential concepts of leadership in a scientific field.
2023	**Workshop KHYS Time-& Self-Management 2023**, presented approaches and tools for improved organization and self-reflexion.
2022	**IBM Neuro-Symbolic AI Summer School**, displayed possible combinations of knowledge-driven, symbolic AI with more traditional data-driven machine learning approaches. Distinguished speakers shared an overview of neuro-symbolic AI and how these method are applied in current applications.
2021	**Workshop Nawik Visualizing Science**, provided insights on how to properly visualize research results to convey the consequent insights.
2021	**Workshop Nawik Communicating Science**, provided insights on how to properly communicate your research results to both experts and layman.